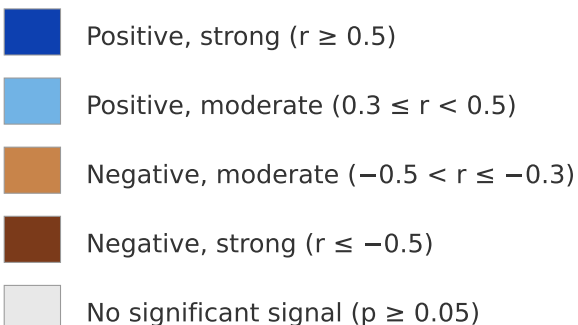


Tonga: how does El Niño generally affect seasonal rainfall?

Pixelwise partial correlation of rainfall with Niño3.4 (ENSO), other climate modes held constant — ERA5 0.25° grid, 1981–2025

Unique ENSO signal — strongest significant trimester per pixel

Partial correlation with Niño3.4

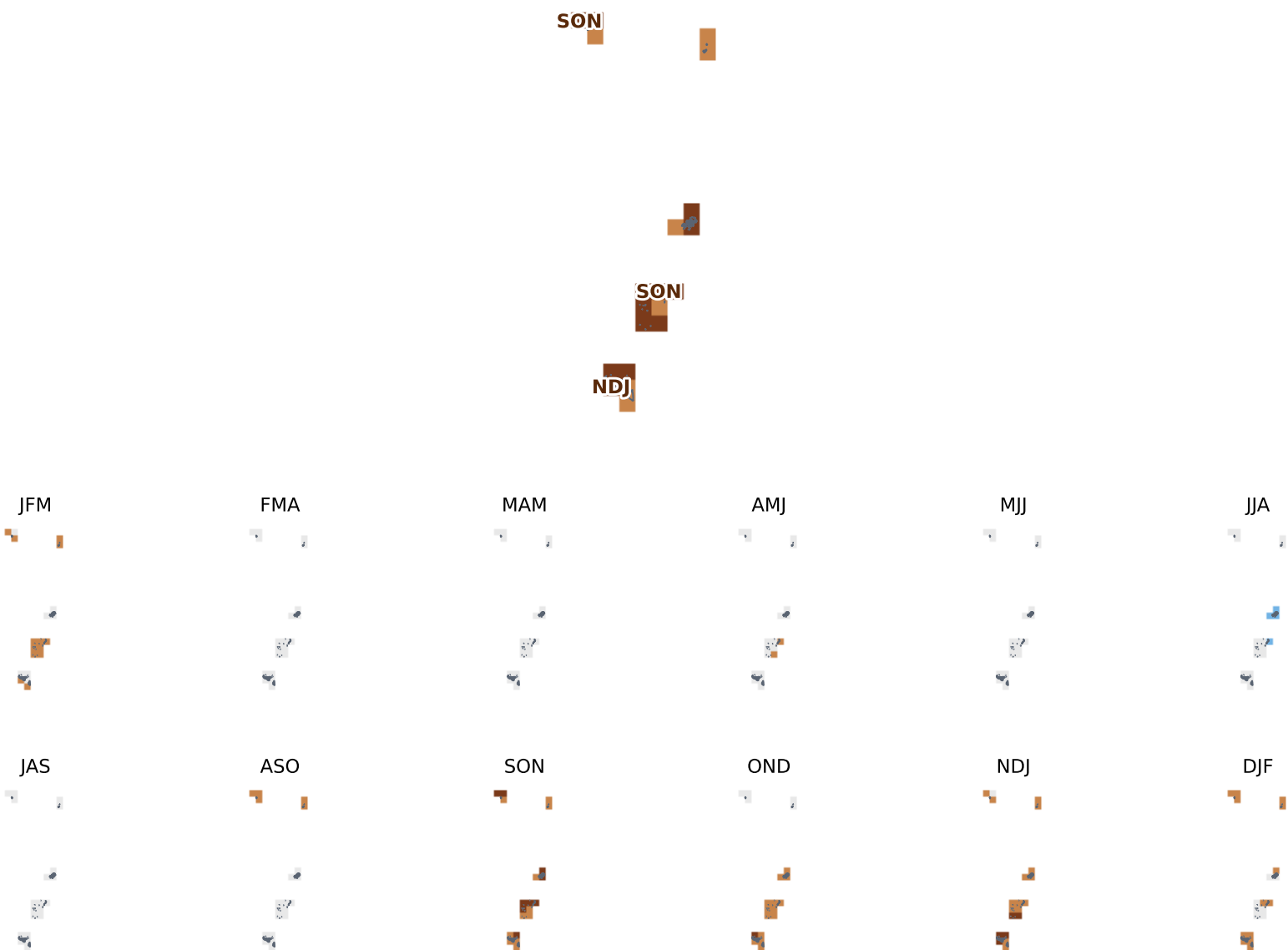


Brown = El Niño years tend to be drier than normal in that season; blue = wetter.

Method (the teleconnections page's partial pass):

- Trimester rainfall means per pixel, all 12 overlapping 3-month windows, 1981–2025
- Partial r of rainfall vs Niño3.4, holding DMI (IOD), TNA, TSA and AMM constant
- Each mode at its per-pixel best lag (0–3 mo)
- Two-tailed $p < 0.05$, df adjusted for controls
- All seasons kept (no rainy-season filter)

Data: ERA5 monthly precip (0.25°), NOAA PSL Niño3.4. Bottom rows: all 12 rolling trimesters.



Tonga: what seasonal rainfall is predicted this year (August-issued forecast)?

SEAS5 (issued August 2026) accounts for El Niño and every other driver · return-period categories vs the 1981–2025 hindcast · skill-shaded · ADM1

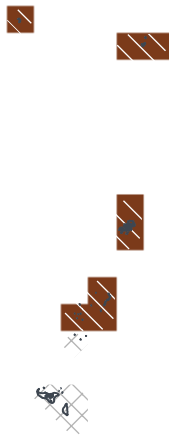
JJA 2026 — in season
1-in-23 dry · 2nd pct



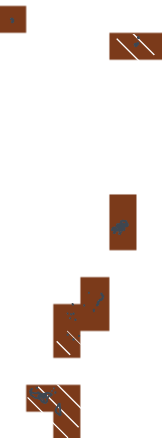
JAS 2026 — in season
1-in-46 dry · 0th pct



ASO 2026 — in season
1-in-46 dry · 0th pct



SON 2026 — lead 1 mo
1-in-46 dry · 0th pct



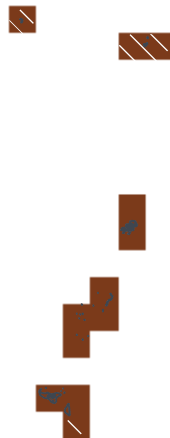
Forecast (high skill):

- Drought — $\geq$ 10-yr return period
- Drought — 3–10-yr return period
- Roughly normal
- Flood — 3–10-yr return period
- Flood — $\geq$ 10-yr return period

Other:

- Mod. skill
- Low skill
- Outside rainy season

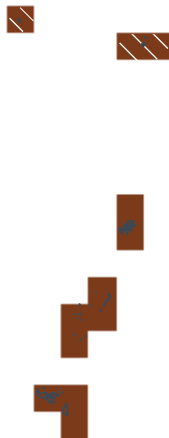
OND 2026 — lead 2 mo
1-in-46 dry · 0th pct



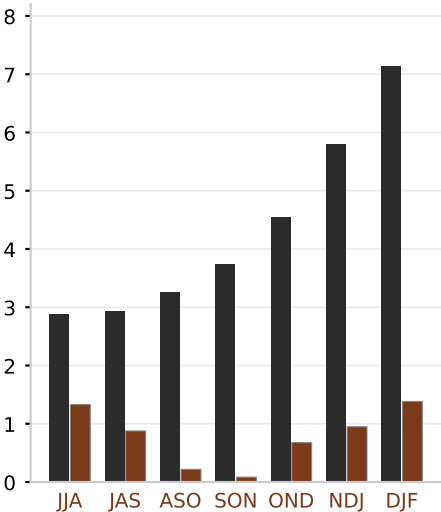
NDJ 2026/27 — lead 3 mo
1-in-46 dry · 0th pct



DJF 2026/27 — lead 4 mo
1-in-46 dry · 0th pct



Country-mean rainfall (mm/day)
black = normal · coloured = forecast



White hatch on a colour = moderate skill.
Tile subtitles: the country-level return period and percentile from the alerts app.
Driest outlook: JAS 2026 (1-in-46 dry).
In season = observed months + forecast.
Niño3.4: +1.7 °C in Jul 2026.